清华大学考试试题专用纸

姓名: _____ 学号: _____

Everyone needs to solve Problem 1, 2, 3, 4 (10 points each). Please select four problems in the following table and solve them (15 points each).

Problem 5	Problem 6	Problem 7	Problem 8	Problem 9	Problem 10

1. (10 points) Let
$$SO(3) = \{A \in M_3(\mathbb{R}) : A^{\mathsf{T}}A = I,\ \det A = 1\}.$$
Compute $H_k(SO(3);\mathbb{Z})$ for every k.

2. Let $M \subset \mathbb{R}^n$ be a compact C^2 embedded submanifold without boundary. Put
$$\nu(M) = \{(p,v) : p \in M,\ v \perp T_pM\},\ \nu_\epsilon(M) = \{(p,v) \in \nu(M) : \|v\| < \epsilon\},$$
and define $\Phi(p,v) = p + v$.

   (a) (5 points) Prove that there exists $\epsilon > 0$ such that $\Phi|_{\nu_\epsilon(M)}$ is a C^1-diffeomorphism onto
$$U_\epsilon(M) = \{x \in \mathbb{R}^n : \mathrm{dist}(x, M) < \epsilon\}.$$

   (b) (5 points) Show that the C^2 hypothesis cannot in general be weakened to C^1. More precisely, let $N \subset \mathbb{R}^2$ be a compact C^1 embedded circle which, in a neighborhood of the origin, is given by the arc
$$\gamma(t) = (t, |t|^{3/2}),\ |t| \le \delta,$$
   You may assume without proof that such an embedded circle N exists. Prove that for every $\epsilon > 0$, the normal map
$$\Phi_N \colon \nu_\epsilon(N) \longrightarrow \mathbb{R}^2,\ \Phi_N(p,v) = p + v,$$
   is not injective.

3. (10 points) Let (M, g) be complete and let X be a Killing field. Prove that every connected component of
$$Z(X) = \{p \in M : X(p) = 0\}$$
is a closed submanifold of even codimension.

4. (10 points) Let M^n be oriented, $n \ge 3$, and let g and h be Riemannian metrics with Hodge-star operators $*_g$ and $*_h$. Suppose
$$*_g\eta = *_h\eta \qquad \text{for every smooth one-form } \eta.$$
Prove that $g = h$ when $n \ge 3$. What happens when $n = 2$?

小红书
小红书号:3189205503 qq

5. Let

$$G = SU(2) = \left\{ \begin{pmatrix} z & w \\ -\overline{w} & \overline{z} \end{pmatrix} : |z|^2 + |w|^2 = 1 \right\}.$$

You may use that G is a compact, connected, smooth real 3-dimensional Lie group. Let

$$\iota \colon G \longrightarrow G,\ \iota(A) = A^{-1},$$

be the inversion map.

(a) (5 points) Determine the Lie algebra $\mathfrak{su}(2) = T_I G$ and prove that

$$d\iota_I = -\operatorname{id}.$$

(b) (10 points) Let $F \colon G \to G$ be a smooth map homotopic to ι. Prove that F has at least one fixed point.

6. Let ∇ and $\widehat{\nabla}$ be torsion-free affine connections on an n-manifold, where $n \geq 2$. Assume that they have the same geodesic paths. In other words, assume that every geodesic for either connection becomes a geodesic for the other after a regular reparametrization.

(a) (7 points) Prove that there is a unique one-form α such that

$$\widehat{\nabla}_X Y = \nabla_X Y + \alpha(X)Y + \alpha(Y)X$$

for all vector fields X, Y.

(b) (8 points) Prove conversely that connections related by this formula have the same geodesic paths.

7. (15 points) Let $X = \mathbb{RP}^2 \vee S^1$, with the wedge point as basepoint. Let a be a loop in $\mathbb{RP}^2$ representing the generator of $\pi_1(\mathbb{RP}^2) \cong \mathbb{Z}/2$, and let b be the standard loop in S^1. Let $f \colon X \to X$ be the based map that is the identity on $\mathbb{RP}^2$ and sends b to $a^{-1}b$. Let

$$T_f = (X \times [0,1])/((x,1) \sim (f(x),0))$$

be the mapping torus of f. Compute $\pi_1(T_f)$ and $H_k(T_f;\mathbb{Z})$ for every k.

8. (15 points) Let (M^n, g) be closed and satisfy $\operatorname{Ric} \geq 0$. Assume that the first Betti number is maximal:

$$b_1(M) = n.$$

Prove that (M, g) is isometric to a flat torus $\mathbb{R}^n/\Lambda$ for some lattice $\Lambda \subset \mathbb{R}^n$.

9. (15 points) Let (M^n, g) be compact and have strictly positive sectional curvature. Let $N^a, P^b \subset M$ be closed embedded totally geodesic submanifolds. Prove that if $a + b \geq n$, then $N \cap P \neq \varnothing$.

10. Let

$$E = \{[X : Y : Z] \in \mathbb{CP}^2 : Y^2 Z = X^3 - XZ^2\}.$$

The set E is connected; you are not required to prove this fact.

小红书
小红书号:3189205503 qq

(a) (5 points) Prove that E is a compact, smooth real 2-dimensional submanifold of $\mathbb{CP}^2$.

(b) (5 points) Equip E with its complex orientation. Define

$$a = \{[x : y : 1] \in E : x \in [-1, 0],\ y \in \mathbb{R}\},\ b = \{[x : y : 1] \in E : x \in [0, 1],\ y \in i\mathbb{R}\}.$$

Show that a and b are smoothly embedded circles intersecting transversely at $p = [0 : 0 : 1]$. Using y as a local complex coordinate at p, orient a and b so that their positive tangent directions at p correspond to -1 and $-i$, respectively. Prove that, with these orientations, $a \cdot b = 1$.

(c) (5 points) It is a fact, which you may use without proof, that E is diffeomorphic to the torus T^2. On the affine chart $Z \neq 0$, write $x = X/Z$ and $y = Y/Z$, and set

$$\omega = \frac{dx}{y}.$$

You may use without proof that ω extends to a smooth closed complex-valued 1-form on all of E. Define

$$\alpha = \operatorname{Re}\omega,\ \beta = \operatorname{Im}\omega,$$

Determine the homology classes

$$\mathrm{PD}([\alpha]),\ \mathrm{PD}([\beta])$$

in $H_1(E; \mathbb{R})$ in terms of $[a]$ and $[b]$. Here, PD denotes the Poincaré dual.

Hint. You may need the following integral:

$$\kappa = 2\int_{-1}^{0} \frac{dx}{\sqrt{x^3 - x}} = 2\int_{0}^{1} \frac{dx}{\sqrt{x - x^3}} = \frac{\sqrt{\pi}\,\Gamma\left(\frac{1}{4}\right)}{\Gamma\left(\frac{3}{4}\right)}.$$

小红书
小红书号:3189205503 qq

Name:_____ Student ID:_____

Qualifying Exam-Analysis and Differential Equations

Fall 2026

1. (10 points) Find all equilibrium points for the system of differential equations for $(x, y) = (x(t), y(t))$,

$$x' = x(1 - x^2 - 6y^2), \quad y' = y(1 - 3x^2 - y^2).$$

and determine their stability in the sense of Lyapunov (asymptotically stable, stable but not asymptotically stable, unstable).

2. (10 points) Is there a one-to-one holomorphic function which maps $\{z \in \mathbb{C} : 0 < |z| < 1\}$ onto the annulus $\{z \in \mathbb{C} : r < |z| < R\}$ where $R > r > 0$? Justify your answer.

3. (10 points) Let $s > 0$, and define the operator D^α as follows

$$\widehat{D^s f}(\xi) = |\xi|^s \widehat{f}(\xi).$$

Show that

$$\|D^s(\varphi u)\|_{L^2(\mathbb{R}^n)} \le C \sup_{|\alpha| \le s+1} \|\partial_x^\alpha \varphi\|_{L^\infty(\mathbb{R}^n)} \|u\|_{H^s}, \quad \forall u, \varphi \in C_c^\infty(\mathbb{R}^n),$$

where C is a universal constant independent of u and φ.

4. (15 points) Let $(X, \mathcal{F}, \mu)$ be a σ-finite measure space and $0 < r < p < +\infty$.

   (a) (10 points) Show that there exists $C = C(p, r, X) > 0$, such that for all measurable function f

$$\frac{1}{C}\|f\|_{L^p_w} \le |||f||| \le C\|f\|_{L^p_w},$$

   Here

$$\|f\|_{L^p_w} := \sup_{\lambda > 0} \lambda[\mu(\{x \in X : |f(x)| > \lambda\})]^{\frac{1}{p}},$$

$$|||f||| := \sup_{\mu(E) < +\infty} \mu(E)^{-\frac{1}{r}+\frac{1}{p}} \left(\int_E |f(x)|^r \, \mathrm{d}\mu(x) \right)^{\frac{1}{r}}$$

小红书 小红书号:3189205503 qq

(b) (5 points) Determine whether the above inequality still holds true if $(X, \mathcal{F}, \mu)$ is not σ-finite.

5. (10 points) For all positive integer m, we denote by

$$\mathcal{Y}_m = \left\{ f \in C^\infty(\mathbb{R}) : \forall k \in \mathbb{N}, \sup_{y\in\mathbb{R}} (1+|y|)^{m+k} |\partial_y^k f(y)| = C_k < +\infty \right\}.$$

Show that $\mathcal{H}f \in \mathcal{Y}_1$ if $f \in \mathcal{Y}_2$. Here $\mathcal{H}f$ is the Hilbert transform of f:

$$\mathcal{H}f(x) = \frac{1}{\pi} \lim_{\varepsilon \searrow 0} \int_{|x-y|>\varepsilon} \frac{f(y)}{x-y} \, \mathrm{d}y.$$

6. (15 points) For $a \in \mathbb{C}$, consider the differential equation for $u = u(x)$

$$xu'' + 2u' + (a-x)u = 0.$$

(a) (5 points) Prove that for every $a \in \mathbb{C}$, there is a nontrivial solution $u \in C^\infty(\mathbb{R})$.

(b) (10 points) Denote $\mathbb{R}_+ = (0, \infty)$, $\mathbb{R}_- = (-\infty, 0)$. Suppose that there is a nontrivial solution $u \in \mathscr{S}'(\mathbb{R})$ (tempered distribution) such that both $u|_{\mathbb{R}_\pm}$ are smooth and both limits $\lim_{x\to 0\pm} u(x)$ exist, show that a must be an even number. (Hint: For example, when $a = 0$, the Dirac δ-function is a solution with the desired properties. In general, consider its Fourier transform $\hat{u}(\xi)$ and the behaviour when $\xi \to \pm\infty$.)

7. (15 points) Let $\Omega \neq \mathbb{C}$ be a simply connected open subset of $\mathbb{C}$. Suppose $f : \Omega \to \Omega$ is holomorphic and f is not the identity function. Prove that f has at most one fixed point in Ω.

8. (15 points) Let $u \in C_c^\infty(\mathbb{R}^2)$ and $A > 100$. Show that

$$\int_{|x|<A} |u(x)|^2 \, \mathrm{d}x \leq CA^2 \log A \int_{\mathbb{R}^2} (|u(x)|^2 e^{-|x|} + |\nabla u(x)|^2) \, \mathrm{d}x.$$

where C is a universal constant independent of u and A.

小红书
小红书号:3189205503 qq

清华大学考试试题专用纸

考试课程 Subject: 求真书院代数博士生资格考 Algebra Qualifying Exam for Qiuzhen College

姓名 Name: _____

学号 Student ID: _____

- Exam Time: September 2026 (3 hours)
- This exam has 4 pages of 7 basic problems and 2 out of 6 advanced problems (100 points).
- Candidates are expected to adhere to the examination discipline by default, and those who do not comply will bear the consequences themselves.
- To receive full credit, all answers must provide necessary **details**, **proofs**, and **justifications**.
- You are strongly encouraged to write your answers **in English**.
- **第 8-13 题为提高题，你需要选择其中任意两题作答。请在答题纸首页清晰画出并填写下列表格，以注明需评分的两道题目。未在首页列出的题目将不予评分。**
  Q.8-13 are advanced problems and **selective**. Choose and answer **two** of them. No other answers will be graded. **Draw and fill the following table on the front page of the answer sheet to declare which two questions need to be graded.**

Advanced Problems		
Q.		
Score		

In the following problems, $\mathbf{Q}$ is the field of rational numbers, $\mathbf{R}$ is the field of real numbers, $\mathbf{C}$ is the field of complex numbers, $\mathbf{Z}$ is the ring of integers, $\mathbf{F}_q$ is the finite field consisting of q elements. **In all the problems, a ring means a commutative ring with multiplicative unit**.

小红书
小红书号:3189205503 qq

1 基础题 Basic Questions

Q. 1 (10 points). *Classify all groups of order 99 up to isomorphism.*

Q. 2 (10 points). *Decide whether every 2×2 complex matrix has a complex matrix square root. Give a proof or a counterexample.*

Q. 3 (10 points). *Let k be a field and let $R = k[x]$. Suppose that $a, b \in R$ are nonzero polynomials, and set*

$$M = R/(a), \qquad N = R/(b).$$

Prove that there exists $d \in R$ such that

$$\mathrm{Tor}_i^R(M,N) \cong \begin{cases} R/(d), & i = 0, 1, \\ 0, & i \geq 2. \end{cases}$$

Deduce that

$$\mathrm{Tor}_1^R(M,N) = 0$$

if and only if d is a unit in R.

Q. 4 (10 points). *Let*

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{pmatrix} \in \mathfrak{gl}_4(\mathbf{C}).$$

Find all Borel subalgebras of $\mathfrak{gl}_4(\mathbf{C})$ that contain A.

Q. 5 (10 points). *Consider the polynomial*

$$f(X) = X^4 - 10X^2 + 20 \in \mathbf{Q}[X].$$

1. *Prove that f is irreducible over $\mathbf{Q}$.*
2. *Prove that the splitting field of f is $L = \mathbf{Q}(\alpha)$ where $\alpha = \sqrt{5 + \sqrt{5}}$.*
3. *Determine $\mathrm{Gal}(L/\mathbf{Q})$, and list all intermediate fields between $\mathbf{Q}$ and L.*

Q. 6 (10 points). *Let*

$$W = \left\langle s_1, s_2 \;\middle|\; s_1^2 = s_2^2 = 1, \quad (s_1 s_2)^4 = 1 \right\rangle.$$

Compute the character table of W.

Q. 7 (10 points). *Let k be a field, and let*

$$R = k[x,y]/(x,y)^2.$$

For each nonzero pair $(a,b) \in k^2$, let $M(a,b)$ be the two-dimensional R-module with k-basis e_1, e_2 whose R-module structure is given by

$$xe_1 = ae_2, \qquad ye_1 = be_2, \qquad xe_2 = ye_2 = 0.$$

小红书
小红书号:3189205503 qq

1. *Prove that $M(a,b)$ is indecomposable.*

2. *Show that the endomorphism ring $\mathrm{End}_R(M(a,b))$ is isomorphic to $k[\varepsilon]/(\varepsilon^2)$ as a k-algebra.*

3. *Let (a,b) and (c,d) be both nonzero pairs in k^2. Prove that*

$$M(a,b) \cong M(c,d)$$

if and only if

$$(c,d) = \lambda(a,b)$$

for some $\lambda \in k^\times$.

4. *Prove that every indecomposable R-module of k-dimension two is isomorphic to $M(a,b)$ for some nonzero pair $(a,b) \in k^2$.*

小红书
小红书号:3189205503 qq

2 高等题 Advanced Questions

Q. 8 (15 points). *Find two sets of closed points* $\{P_1, P_2, P_3\}, \{Q_1, Q_2, Q_3\} \subset \mathbf{A}^1$ *such that*

$$\mathbf{A}^1 \setminus \{P_1, P_2, P_3\} \ncong \mathbf{A}^1 \setminus \{Q_1, Q_2, Q_3\}.$$

(Assume the base field is $\mathbf{C}$*.)*

Q. 9 (15 points). *Determine whether the following projective plane curves are smooth, and compute the geometric genus of their normalizations. (Assume char* $k = 0$ *or char* $k \geq 5$*).*

1. $YZ = X^2$.
2. $Y^2Z = X^3$.
3. $Y^2Z = X^3 - XZ^2$.

Q. 10 (15 points). *Let* A *be a Dedekind domain. Denote by* K *the fraction field of* A*.*

1. *Let* $\mathfrak{p}_1, \cdots, \mathfrak{p}_n$ *be pairwise distinct maximal ideals of* A*. Let* $e_1, \cdots, e_n$ *be elements of* $\mathbf{Z}$*. Show that there exists* $x \in K$ *such that in the factorization of the fractional ideal* (x)*, the exponent of* $\mathfrak{p}_i$ *is* e_i *for all* $1 \leq i \leq n$*.*
2. *Show that any ideal of* A *can be generated by two elements.*

Q. 11 (15 points). *Let* ζ_7 *be a* 7*-th primitive root of* 1 *and* $K = \mathbf{Q}(\zeta_7)$*. Let* $q \in \mathbf{Z}$ *be a prime number and let* β *be a maximal ideal of* $\mathcal{O}_K$ *above* q*. Determine the residue degree* $f_{\beta/q}$*.*

Q. 12 (15 points). *Let* $\mathfrak{g}$ *be a finite-dimensional semisimple complex Lie algebra. Fix a Cartan subalgebra and a Borel subalgebra of* $\mathfrak{g}$*. Let* $\mathfrak{n}$ *and* $\mathfrak{n}^-$ *respectively be the subalgebra spanned by positive and negative roots. Let* $Z(U\mathfrak{g})$ *be the center of the enveloping algebra* $U\mathfrak{g}$*. Show that*

$$Z(U\mathfrak{g}) \cap (U\mathfrak{n} \cdot U\mathfrak{n}^-) = \mathbf{C} \cdot 1.$$

Q. 13 (15 points). *Let* $\mathcal{O}_0$ *be the principal block of the BGG category* $\mathcal{O}$ *for* $\mathfrak{g} = \mathfrak{sl}_3$*.*

(1) Find the highest weights of all the indecomposable projective modules in $\mathcal{O}_0$*.*

(2) For each module in (1), find its length.

You may use the following fact: each Kazhdan–Lusztig polynomial for S_3 *is either zero or a monomial with coefficient* 1*.*

小红书
小红书号:3189205503 qq